ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data

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***Abstract*—Lifelog data enable non-invasive prediction of everyday health states such as sleep quality, fatigue, and stress, but reliable modeling remains challenging due to high dimensionality, missingness, and inter-individual variability. GBDT models effectively learn structured numerical relationships, whereas LLMs provide a complementary way to represent personalized behavioral and temporal context, though they may be less reliable on small-scale datasets.**

We propose *ConFuse*, a context-aware LLM–GBDT fusion framework for sleep state prediction. The GBDT branch combines LightGBM, XGBoost, and CatBoost through target-wise weighted blending, while the LLM branch predicts from a *personal-baseline-aligned context* using binary-token fine-tuning and target-wise Platt calibration. A GPT-4.1-based *Data Reliability Score* adaptively controls the sample-wise contribution of the LLM during fusion.

On the 90 development-validation cases of the ETRI lifelog dataset, ConFuse achieved a mean binary log loss of 0.6415, outperforming the GBDT ensemble (0.6530), the context-aware LLM (0.6735), and a fixed 50:50 fusion (0.6445).

I. Introduction

Sleep quality, fatigue, and stress are closely tied to physical and mental health [1], and wearables continuously capture lifelog signals (screen usage, activity, mobility, heart rate) for non-invasive sleep-state prediction [2].

GBDT-based models handle high-dimensional features and missing values well on structured lifelog data, but identical sensor readings can reflect different sleep contexts depending on habitual behavior. This has motivated recent work to serialize tabular and physiological data into natural-language prompts for LLMs, which add personalized and temporal context but remain less reliable for numerical reasoning on small datasets and are typically fused with fixed weights regardless of per-sample quality.

To address these limitations, we propose **ConFuse**, a context-aware LLM–GBDT fusion framework combining numerical lifelog patterns with personalized behavioral context through: (1) a *personal-baseline-aligned context* with binary-token fine-tuning that directly optimizes the logit difference between tokens `0` and `1`; (2) target-wise blending of LightGBM, XGBoost, and CatBoost; and (3) a reliability-guided fusion strategy that adaptively weights LLM predictions via a GPT-4.1-based *Data Reliability Score*.

II. Related Works

A. Lifelog-Based Sleep Prediction

Lifelog-based sleep prediction has followed two main directions: machine learning with engineered statistical features, or deep learning on temporal sensor sequences. Representative studies include LSTM-based multimodal modeling [3], subject-adaptive sparse linear models [4], and feature-engineered GBDT ensembles for the ETRI Human Understanding AI Challenge dataset [5]. These approaches do not jointly model personal historical baselines and reliability-guided model fusion. ConFuse combines target-wise GBDT blending with individual-history features summarizing each subject's behavioral and target patterns.

B. LLMs for Tabular and Wearable Data

Numerical and wearable data can be processed by LLMs through natural-language serialization, including TabLLM for tabular classification [6], Health-LLM for wearable-based health prediction [7], PH-LLM for personal health modeling [8], and PhysioLLM for physiological insights [9]. These studies explore LLM-based prediction from tabular or wearable data, whereas ConFuse combines LLM and GBDT predictions through reliability-guided fusion. ConFuse preserves the numerical modeling capability of GBDTs while using the LLM for personalized behavioral context, with its contribution weighted by the reliability of each lifelog sample.

III. Methods

A. Overview of ConFuse

Fig. 1 illustrates the overall architecture of ConFuse. Given a subject–date lifelog, the GBDT ensemble branch and the context-aware LLM branch (the latter fine-tuned via supervised fine-tuning, SFT) predict probabilities for the seven sleep targets in parallel. The GBDT branch uses engineered numerical features with target-wise weighted blending, while the LLM branch predicts from a personal-baseline-aligned context. A GPT-4.1-based LLM Reliability Score is then converted into a sample-wise fusion weight to combine the two predictions.

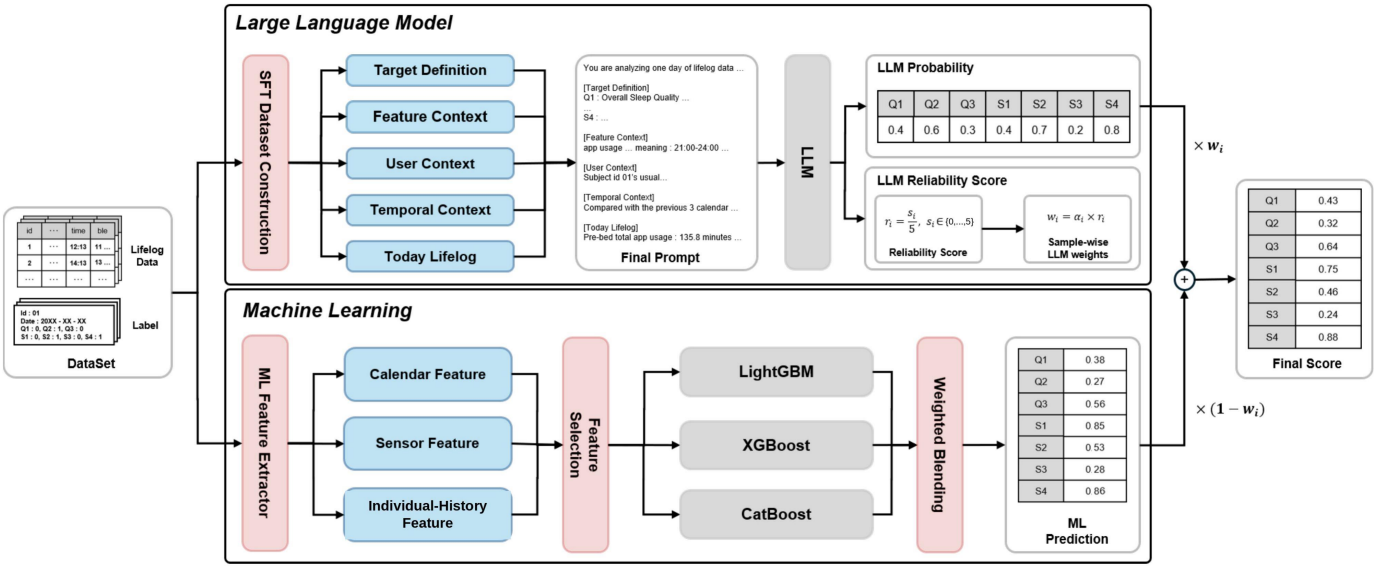


Fig. 1. **Overview of ConFuse.** GBDT and context-aware LLM predictions are combined using a GPT-4.1-derived reliability weight.

B. GBDT Ensemble Branch

1) Feature Construction: The ML branch uses 2,449 sensor, calendar, and individual-history features. Sensor features summarize sensor streams using descriptive and temporal statistics, whereas individual-history features consist of rolling Q1–Q3 statistics and subject-level persona/reactivity statistics.

2) Target-Specific Feature Selection: Because the number of features far exceeds the number of training cases, a low correlation threshold ($|r| \geq 0.02$) was used to retain weak signals, and the top 500 features were kept per target by absolute Pearson correlation.

3) Model Training and Weighted Blending: Five-fold cross-validation over per-subject date blocks on the 360 training cases gave a stable out-of-fold (OOF) estimate for the subsequent weight search while keeping each fold inexpensive to retrain. LightGBM, XGBoost, and CatBoost were trained independently on the same folds and selected features; target-averaged OOF log losses were 0.5863, 0.6047, and 0.5840, respectively. Following a data-driven weighted-ensemble approach [10], we performed a grid search over target-specific blending weights $\lambda_{m,t}$ using training out-of-fold predictions. The final ML probability is

$$\hat{p}_{\mathrm{ML}}(i,t) = \sum_{m \in \mathcal{M}} \lambda_{m,t}\, \hat{p}_m(i,t), \tag{1}$$

where $\mathcal{M} = \{\mathrm{LightGBM}, \mathrm{XGBoost}, \mathrm{CatBoost}\}$, $\lambda_{m,t} \geq 0$, and $\sum_m \lambda_{m,t} = 1$. The target-wise weighted blending achieved a mean out-of-fold log loss of 0.5796, lower than that of any single model. Applying the weights determined on the training data, held fixed, to the 90 development-validation cases, the GBDT ensemble recorded a mean log loss of 0.6530.

C. Context-Aware LLM Branch

1) LLM-guided Feature Selection: Features for the LLM branch were selected independently of the ML branch: following the LLM-Score strategy of LLM-Select [11], GPT-4.1 ranked 169 candidate features by expected usefulness for sleep prediction using only semantic descriptions and data-quality metadata (no label or feature–label correlation access); after redundancy removal and adding one physiological feature, 21 features remained.

2) Personal-Baseline-Aligned Context: The LLM input consists of three sections: `[CASE]`, `[ALIGNED_LIFELOG]`, and `[RECENT_3D_SUMMARY]` (Fig. 2).

`[ALIGNED_LIFELOG]` aligns each observation with the subject's historical baseline, whereas `[RECENT_3D_SUMMARY]` summarizes behavioral trends over the previous three days.

3) Binary-Token Fine-Tuning: Qwen2.5-7B-Instruct [12] was fine-tuned with 4-bit QLoRA (NormalFloat and double quantization [13]). For each subject–date case, seven target-specific QA samples (Q1–Q3, S1–S4) were generated. Three-fold cross-validation produced out-of-fold predictions, with all seven QA samples from each subject–date case kept in the same fold. A new LoRA adapter was trained from the same pretrained backbone for each fold.

At the first answer position for case i and target t, the logit difference between the vocabulary logits corresponding to `1` and `0` is defined as

$$z_{it} = \ell_{it}(1) - \ell_{it}(0). \tag{2}$$

Without a separate classification head, $\sigma(z_{it})$ is used as the raw class probability. With label smoothing $\epsilon = 0.1$ and the smoothed target $\widetilde{y}_{it} = (1-\epsilon)y_{it} + \epsilon/2$, the binary-token loss is computed as

$$\mathcal{L}_{\mathrm{BT}} = \frac{1}{|\mathcal{D}|} \sum_{(i,t)\in\mathcal{D}} \mathrm{BCEWithLogits}(z_{it}, \widetilde{y}_{it}), \tag{3}$$

where $\mathcal{D}$ denotes the set of target-specific QA samples. Historical statistics were computed only when at least three prior

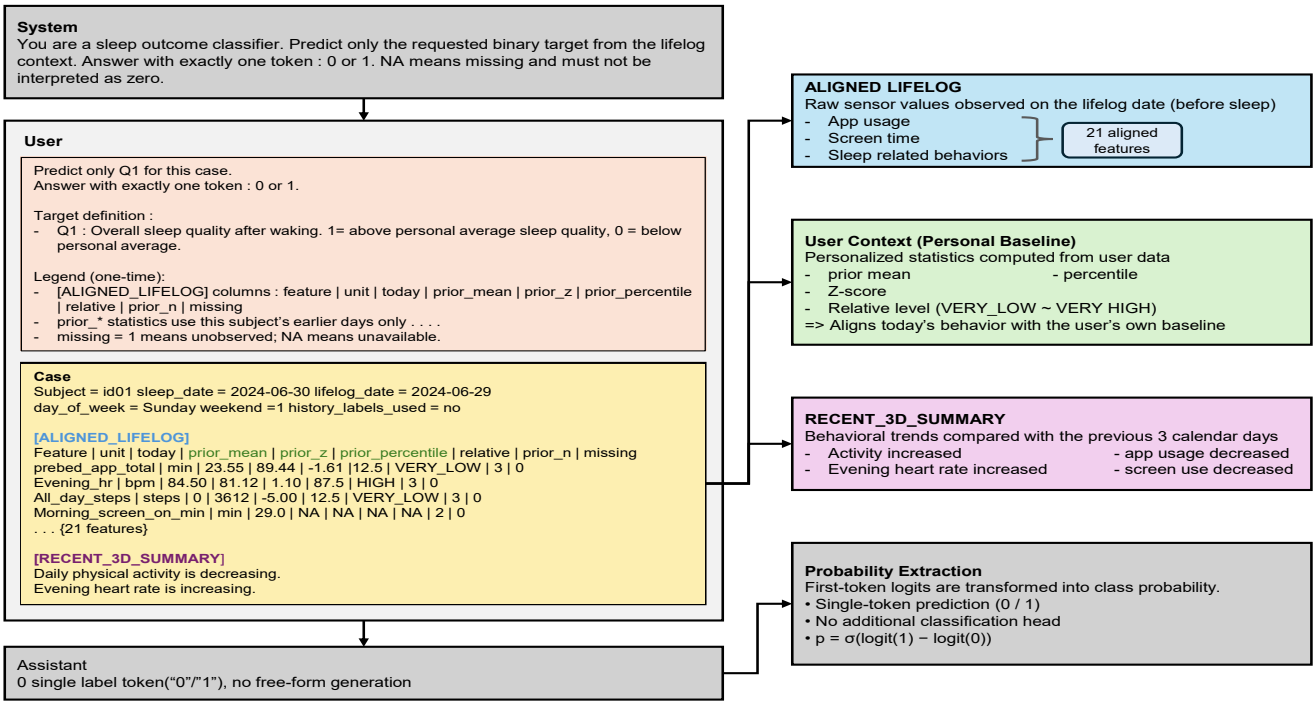


Fig. 2. Context-aware LLM prompt for supervised fine-tuning

valid observations were available; z-scores were clipped to $[-5, 5]$, and insufficient or degenerate histories were marked as NA. No sleep labels are used to construct these contexts.

4) Probability Calibration: Target-wise Platt calibration was fitted on the training out-of-fold logits to improve probability calibration. For the development-validation set, calibration was applied to the logits of the three adapters and the resulting probabilities were averaged:

$$\hat{p}_{it}^{\mathrm{cal}} = \frac{1}{3} \sum_{f=1}^{3} \sigma\left(a_t z_{it}^{(f)} + c_t\right), \tag{4}$$

where a_t and c_t are the calibration parameters for target t. The calibrated probabilities were used for both standalone LLM evaluation and the final fusion.

D. GPT-4.1-Based Data Reliability Assessment

GPT-4.1 assigns multiple semantic scores to each subject–date case, including a Data Reliability Score $s_i \in \{0, \ldots, 5\}$ based on information sufficiency, missingness, baseline availability, and signal consistency. Only the Data Reliability Score is used for fusion, without predicting sleep labels or using ground-truth outcomes. The score is normalized to determine the sample-wise LLM weight.

E. Reliability-Guided Fusion

The GBDT ensemble learns statistical relationships among structured numerical features, whereas the context-aware LLM processes behavioral meaning from natural-language representations that include the personal baseline and recent changes. Because the quality of the contextual information available to the LLM differs from case to case, ConFuse does not apply the same LLM weight to all samples. Normalizing the Data Reliability Score as $q_i = s_i/5 \in [0, 1]$, the per-sample LLM weight is computed as

$$w_i = \alpha q_i, \tag{5}$$

where α is a constant that caps the maximum contribution of the LLM. In this work α was fixed to 0.45 before fusion performance was evaluated, so that the GBDT ensemble retains the majority of the weight even for the most reliable cases. The final ConFuse probability for case i and target t is

$$\hat{p}_{\mathrm{ConFuse}}(i,t) = (1 - w_i)\, \hat{p}_{\mathrm{ML}}(i,t) + w_i\, \hat{p}_{\mathrm{LLM}}^{\mathrm{cal}}(i,t). \tag{6}$$

Higher reliability increases the LLM contribution without introducing a separate trainable fusion classifier.

IV. Experimental Setup

A. Dataset and Prediction Targets

We use the ETRI Human Understanding AI Challenge lifelog data [14]: 12 smartphone/smartwatch sensor logs from ten participants (id01–id10, June 4–Nov 15, 2024; 450 subject–date cases), split into 360 training and 90 development-validation cases by a stratified random split on `subject_id`. The seven binary targets are Q1–Q3 (self-reported sleep quality, fatigue, and stress; binarized relative to each participant's mean) and S1–S4 (sleep time, efficiency, onset latency, and wake after onset from the Withings Sleep Analyzer; binarized per National Sleep Foundation recommendations [15], [16]).

TABLE I
SINGLE-SEED BACKBONE COMPARISON

Model	OOF Raw ↓	OOF AUROC ↑	Dev. Raw ↓	Dev. + Platt ↓
Qwen2.5-7B-Instruct	0.6603	0.5580	0.6782	**0.6735**
Qwen2.5-14B-Instruct	0.6633	**0.5693**	0.6784	0.6757
Qwen3-8B	**0.6598**	0.5494	0.6779	0.6795
Llama-3.1-8B-Instruct	0.6612	0.5521	0.6805	0.6770
Phi-4	0.6610	0.5515	**0.6758**	0.6744
Qwen3-14B	0.6713	0.5377	0.6933	0.6771

AUROC: Area Under the ROC Curve

TABLE II
COMPARISON OF STANDALONE AND FUSION METHODS

Method	Mean Log Loss ↓	Δ vs. GBDT (+ = better)
GBDT Ensemble	0.6530	—
Context-Aware LLM	0.6735	-0.0205
Fixed LLM–GBDT Fusion	0.6445	$+0.0085$
ConFuse	**0.6415**	$\mathbf{+0.0115}$

B. Evaluation Metric

Performance was evaluated using the official binary log loss. Letting y_{it} and $\hat{p}_{it}$ denote the ground-truth label and predicted probability for case i and target t, respectively, the loss is $\ell_{it} = -y_{it} \log \hat{p}_{it} - (1 - y_{it}) \log(1 - \hat{p}_{it})$. The final score is obtained by averaging the log loss over all 90 development-validation cases and seven targets.

C. Implementation Details

Qwen2.5-7B-Instruct was fine-tuned using 4-bit QLoRA with NormalFloat and double quantization, LoRA rank 16, alpha 32, and dropout 0.05. We trained each adapter for three epochs with a learning rate of 1×10^{-4}, an effective batch size of 16, and a maximum sequence length of 1,280 using bf16 and paged 8-bit AdamW. Training and fold-assignment seeds were set to 42. A single-seed comparison of six instruction-tuned backbones was conducted; since the development-validation set was used for backbone selection, the results are not fully independent test-set estimates.

V. RESULTS AND DISCUSSION

A. LLM Backbone Comparison

After Platt calibration, Qwen2.5-7B-Instruct achieved the lowest development-validation log loss (0.6735) among the six backbones (Table I). Considering predictive performance together with computational efficiency, it was selected as the backbone of ConFuse.

B. Data Reliability Distribution

On the development-validation set, GPT-4.1 assigned Data Reliability Scores across the full range 0–5, with a mean of 3.60 (average LLM weight 0.324), enabling sample-wise weighting rather than a fixed fusion weight.

C. Fusion Performance

The standalone GBDT ensemble outperformed the context-aware LLM (Table II), indicating that tree-based models better captured structured numerical features under limited training data, while the fixed LLM–GBDT fusion still beat both standalone models, suggesting the two branches carry complementary information.

ConFuse achieved the lowest mean log loss (0.6415); although its gain over fixed fusion was modest, reliability-guided fusion adaptively adjusts the LLM's contribution without an additional trainable fusion model.

VI. CONCLUSION

In this study, we proposed ConFuse, a context-aware LLM–GBDT fusion framework combining GBDTs' numerical modeling with an LLM's personalized contextual reasoning through reliability-guided fusion, adaptively weighting the LLM's contribution by input quality rather than a fixed strategy. In the development-validation experiments, ConFuse achieved the lowest mean log loss (0.6415), outperforming the standalone GBDT ensemble, the standalone LLM, and a fixed 50:50 fusion baseline. Future work includes validation on larger cohorts and unseen subjects to assess generalizability.

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